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DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY



A PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS

ON

"DETECTION OF CIPHER TYPES USING ARTIFICIAL NEURAL NETWORKS"

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Certificate

It is certified hereby that this report, titled "Detection of Cipher Types using Artificial Neural Networks", and all the attached documents herewith are authentic records of Rajdeep Ghosh (510519002), Swarnali Mondal (510510103), Krishnendu Bera (510519004), Arnab Sen (510519006), Sanjana Chakravarty (510519038), and Abhiroop Mukherjee (510519109) from the Prestigious Department of Computer Science And Technology of the Distinguished and Respected IIEST Shibpur under my guidance.

The works of these students are satisfies all the requirements for which it is submitted. To the extent of my knowledge, it has not been submitted to any different institutions for the awards of degree.

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1 INTRODUCTION

Information is the power in the present day. Information and data should be protected from unauthorized access, distortion, or any unauthorized activity.

Cryptography is the study of secret writing. Information is no longer hidden due to the existence of network adversaries. Our communication channel is completely unstable; a third party might covertly eavesdrop and obtain information from the network or from the channel. Here, cryptography is crucial in preventing unauthorized access to our data. Using key(s), cryptography converts plaintext messages into unreadable ciphertext and vice versa. Encryption is the process of employing a key(s) to transform the original message into an unintelligible format, while decryption is the act of using a key(s) to convert the cipher text back to plaintext. The messages can only be encrypted or decrypted by the authorized user.

The project's objective is to examine encrypted text files to find and categorize their cipher types using artificial neural networks. The cipher algorithm classification model is based on the steps:

1. Creation of a collection of cipher texts with various ' cipher texts.
2. Building a Deep Neural Network (DNN) and training it with a variety of cipher texts and training parameters to enable accurate learning and cipher text classification.
3. Determine the cipher text's kind.

First, the Caesar, Affine, Vigenère, Substitution, and Rail Fence ciphers were used to encrypt English texts. Following the submission of the encrypted files to the CNN and Transformer models, predictions are created, and accuracy is assessed.

2 PRELIMINARIES

To ramp up for this project, we needed a basic knowledge of neural networks and deep learning and an overview of the encryption algorithms that we would be targeting in this project.

Having some knowledge of encryption algorithms might be useful for feature extraction.

The ML topics that were covered in this project are:

- Perceptron
- Artificial Neural Network (ANN)
- Recurrent Neural Network (RNN)
- Convolutional Neural Network (CNN)
- Transformers
- Attention Mechanism in Deep Learning

3 LITERATURE SURVEY

4 THE PROJECT

4.1 Software Used

The softwares used to build this *checker* are:

4.1.1 An Integrated Development Environment (IDE)

An integrated development environment (IDE) is a software application that provides comprehensive facilities to computer programmers for software development. An IDE normally consists of at least a source code editor, build automation tools and a debugger. Some IDEs contain the necessary compiler, interpreter, or both; others, do not.

We used PyCharm as our IDE, as it was easy to set up and code

wikipedia referencePyCharm is an integrated development environment (IDE) used in computer programming, specifically for the Python language. It is developed by the Czech company JetBrains. It provides code analysis, a graphical debugger, an integrated unit tester, integration with version control systems (VCSes), and supports web development with Django as well as data science with Anaconda.

4.1.2 Python

Python is an interpreted, high-level and general-purpose programming language. Python's design philosophy emphasizes code readability with its notable use of significant whitespace. Its language constructs and object-oriented approach aim to help programmers write clear, logical code for small and large-scale projects.

Why did we choose Python?

1. Python has an upper hand when it comes to software based on image recognition and object detection. Since it is the main objective of the project, choosing python was a given.Python has an upper hand when it comes to software based on image recognition and object detection. Since it is the main objective of the project, choosing python was a given.
2. Python is unbeaten when it comes to Machine Learning. Python has support for myriad machine learning libraries, such as OpenCV, the one being used here.
3. Python is comparatively easier to understand and learn. The syntax is clear and simple to read and write.
4. And just our overall experience of using python for years.

4.1.3 Google Colab

After working on the project for quite some time, we realized that we did not have enough hardware resources at out disposal to actually make the *checker* work smoothly. So we decided on shifting to Google Colab. Google colab is an online iPython development environment similar to Jupyter Notebook. It uses CUDA acceleration to speed up processes, so we switched to it rather than continuing development

4.1.4 LaTeX

LaTeX was used to write this report. LaTeX is a software system for document preparation. When writing, the writer uses plain text as opposed to the formatted text found in "What You See Is What You Get" word processors like Microsoft Word or LibreOffice Writer.

4.2 The Program

4.2.1 Outline

The blueprint of this *checker* that we thought of initially:

1. Video Input

Need some way to handle video input coming through the camera feed

2. Processing

   The input needs to be processed somehow

3. Detecting people

   Need to identify people in the video feed

4. Measuring distance between each couple

   Need to calculate the distance between every two persons

5. Mark the violations

   Need to mark the ones that violate social distancing norms

4.2.2 Proceedings

How we proceeded with the outlines of the blueprint:

1. Video Input

   This was easier than we expected it to be. We just had to get our hands on some recorded footage of somewhat populated areas. We refrained from using live footage because:
   - It is tough to get our hands on the light footage of a security camera or the equivalent.
   - If the checker worked on recorded footage, it would work on live footage as well.

The videos we ended up choosing:



(a) Pedestrian Video (b) Shibuya Video

Figure 1: Still Pictures from Sample Videos

2. Processing

We used the OpenCV library for our video/image processing. It is a really handy library that can be used for image processing, object detection and many other purposes.

```
import cv2 as cv

def total_frames(file_name):
    cap = cv.VideoCapture(file_name)
    res = 0

    while True:
        ret,img = cap.read()

        if not ret:
            break

        res = res+1

    return res

file_name = "pedestrian.mp4"
tot_frame = total_frames(file_name)
print(f"Total Frames in {file_name} are {tot_frame}")
```

Listing 1: A Sample Code To Count No. Of Frames in a Video

3. Detecting People

- For this we decided to go with the You Only Look Once (YOLO) algorithm for object detection. The algorithm itself is discussed a bit later in the report.
- We did not train the object detection neural network model ourselves. We used the prebuilt model, trained by the Darknet team https://pjreddie.com/darknet/yolo/ because of time constraints.

4. Measuring distance between each couple

- This was undeniably the toughest part of the project and took the longest time. First we decided to go with measuring the distance between the centroids of every two detections. But that may not work in every condition since it depends on the placement of camera and the view angle from the ground and perpendicular to the ground.
- A conversion of the 3-dimensional footage being fed to the algorithm to 2-dimensions was more than necessary to get the top view of every frame to avoid the *viewing angle problem*.
- Enter **Bird's Eye View (BEV)**. This is what we called the top view of every frame. This was made possible by OpenCV's getPerspectiveTransform() and warpPerspective() functions.

```
def birds_eye_view(corner_points,width,height,image):
    """
    Compute the transformation matrix
    corner_points : 4 corner points selected from the image
    height, width : size of the image
    return : transformation matrix and the transformed image
    """
    # Create an array out of the 4 corner points
    corner_points = np.float32(corner_points)
    # Create an array with the parameters (the dimensions) required to build the matrix
    img_params = np.float32([[0,0],[width,0],[0,height],[width,height]])
    # Compute and return the transformation matrix
    matrix = cv.getPerspectiveTransform(corner_points,img_params)
    img_transformed = cv.warpPerspective(image,matrix,(width,height))

    return matrix,img_transformed
```

Listing 2: Function Bird's Eye Perspective Transformation Matrix

- homography estimation pdf This piece of code essentially calculates what is called a *transformation matrix* for the supplied image (frame) which can then be used to get the centroids of the points as seen from a vertical position directly above the center of the rectangle passed to the function.

```
def birds_eye_point(matrix,centroids):
    """ Apply the perspective transformation to every ground point which have been detected
     on the main frame.
    @ matrix : the 3x3 matrix
    @ centroids : list that contains the points to transform
    return : list containing all the new points
    """
    # Compute the new coordinates of our points
    points = np.float32(centroids).reshape(-1, 1, 2)
    transformed_points = cv.perspectiveTransform(points, matrix)
    # Loop over the points and add them to the list that will be returned
    transformed_points_list = list()

    for i in range(0,transformed_points.shape[0]):
        transformed_points_list.append([transformed_points[i][0][0],transformed_points[i
    ][0][1]])

    return transformed_points_list
```

Listing 3: Function which convert co-ordinates to it's bird's view co-ordinate

- We used these functions to get a two dimensional view of every frame and calculate distance between every pair of detections (people).

5. Mark the violations

   This was again a fairly easy step. We just needed the coordinates of the people in the *violation zone* and make their detection rectangle red as opposed to green.

4.3 The YOLO Algorithm